INTPART summer school, 2025

Your name goes here

January 11, 2026

Abstract

A short overview of what your report is about.

1 Introduction

2 Nuclear physics and nuclear astrophysics

1. How you add bullets
2. Add a new bullet

2.1 Elements and their production in the Solar System

2.2 Nuclear Burning stages of stars

2.3 The production of heavy nuclei

s-process

r-process

p-process

Table 1: Table showing the typical energy ranges for the different resonances

resonance	Typical energy range (MeV)
GDR	

3 Nuclear Level Density and Gamma Strength Function

4 TALYS

5 TALYS exercises

5.1 Exercise 1

5.1.1 ^{191}Os

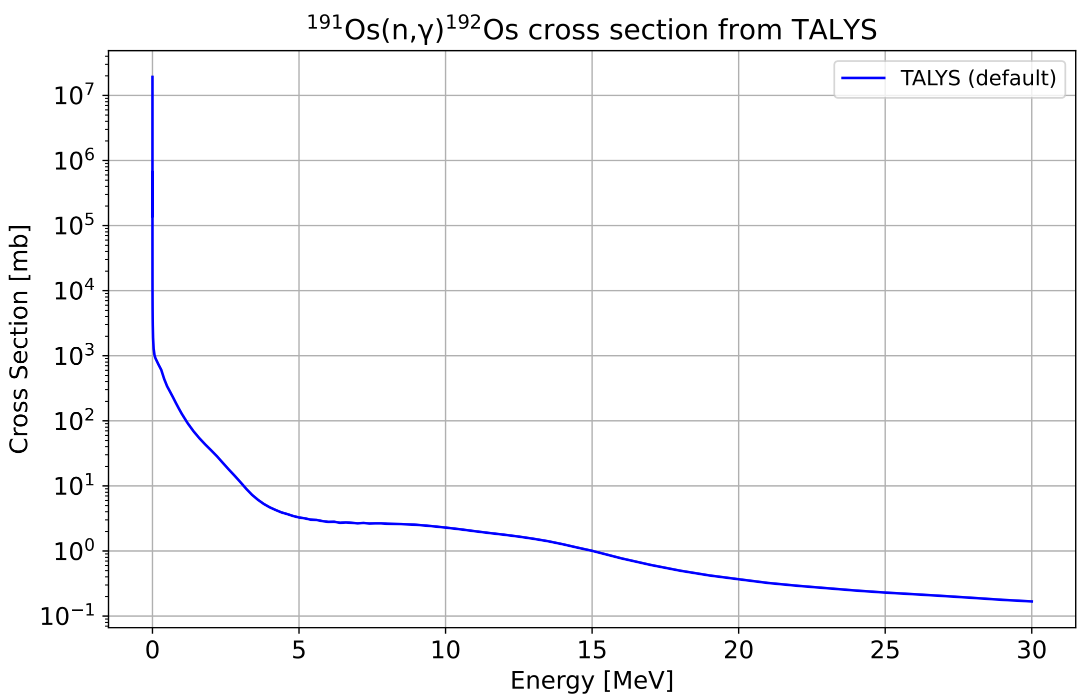

Figure 1: Caption

(n, γ)

(n, n')

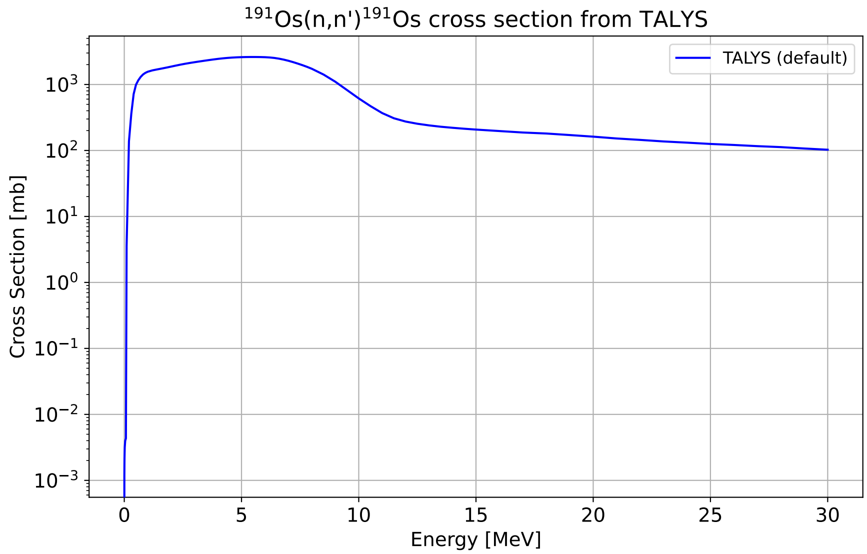

Figure 2: Caption

5.1.2 ^{191}Re

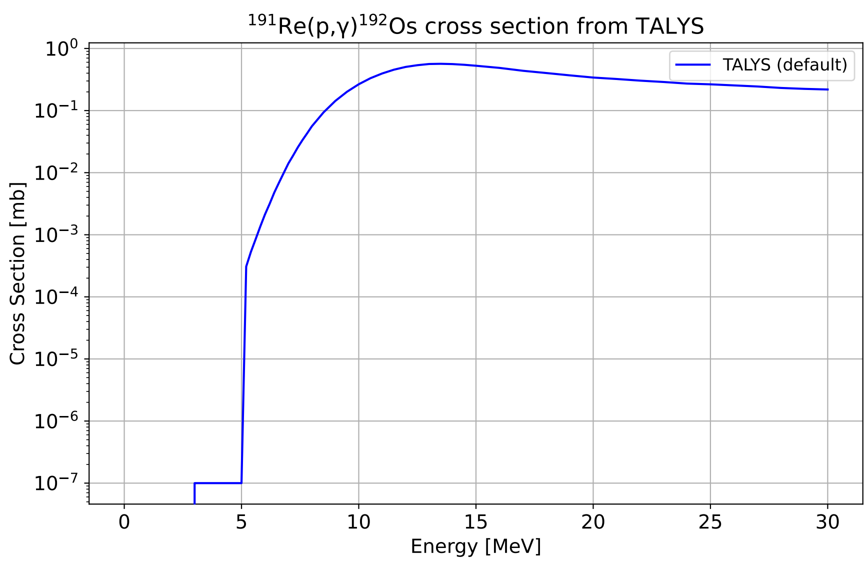

Figure 3: Caption

(n, γ)

5.2 Exercise 2

5.2.1 ^{208}Pb

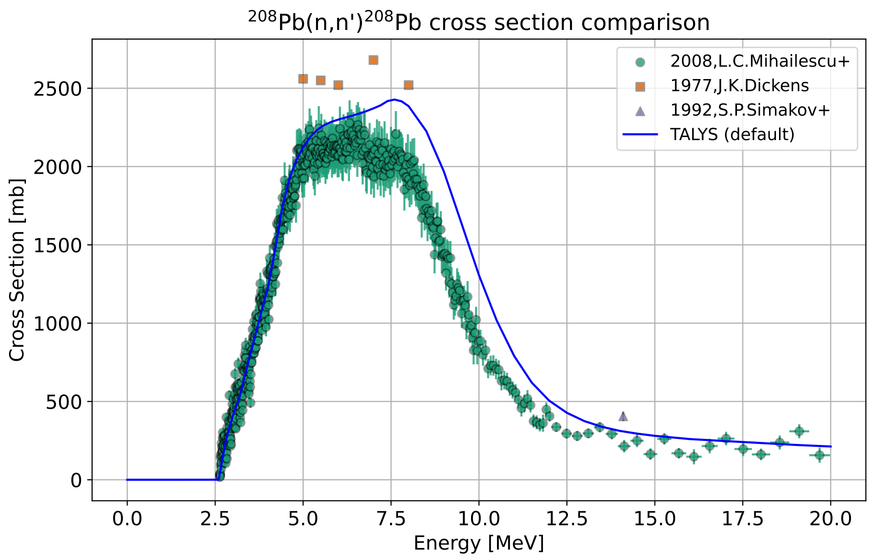

Figure 4: Caption

(n, n')

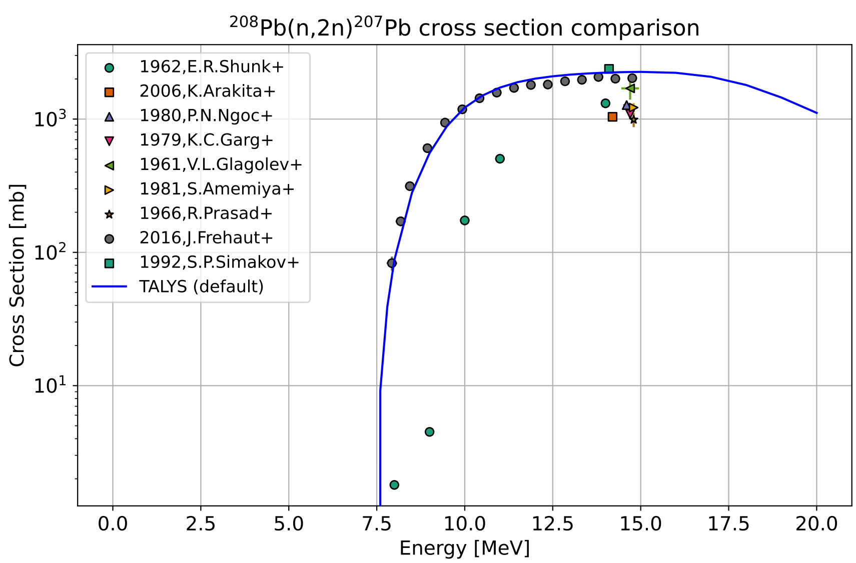

Figure 5: Caption

$(n, 2n)$

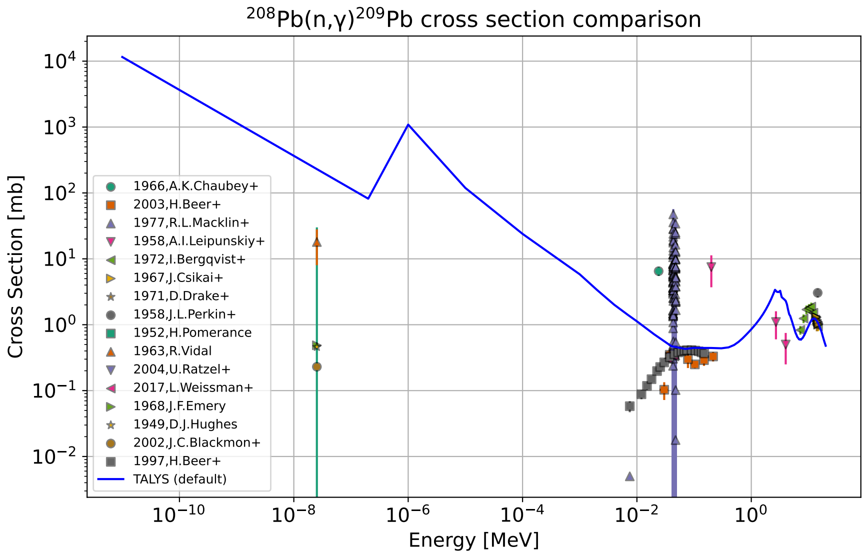

Figure 6: Caption

(n, γ)

5.3 Exercise 3

5.3.1 ^{208}Pb With Varying Models

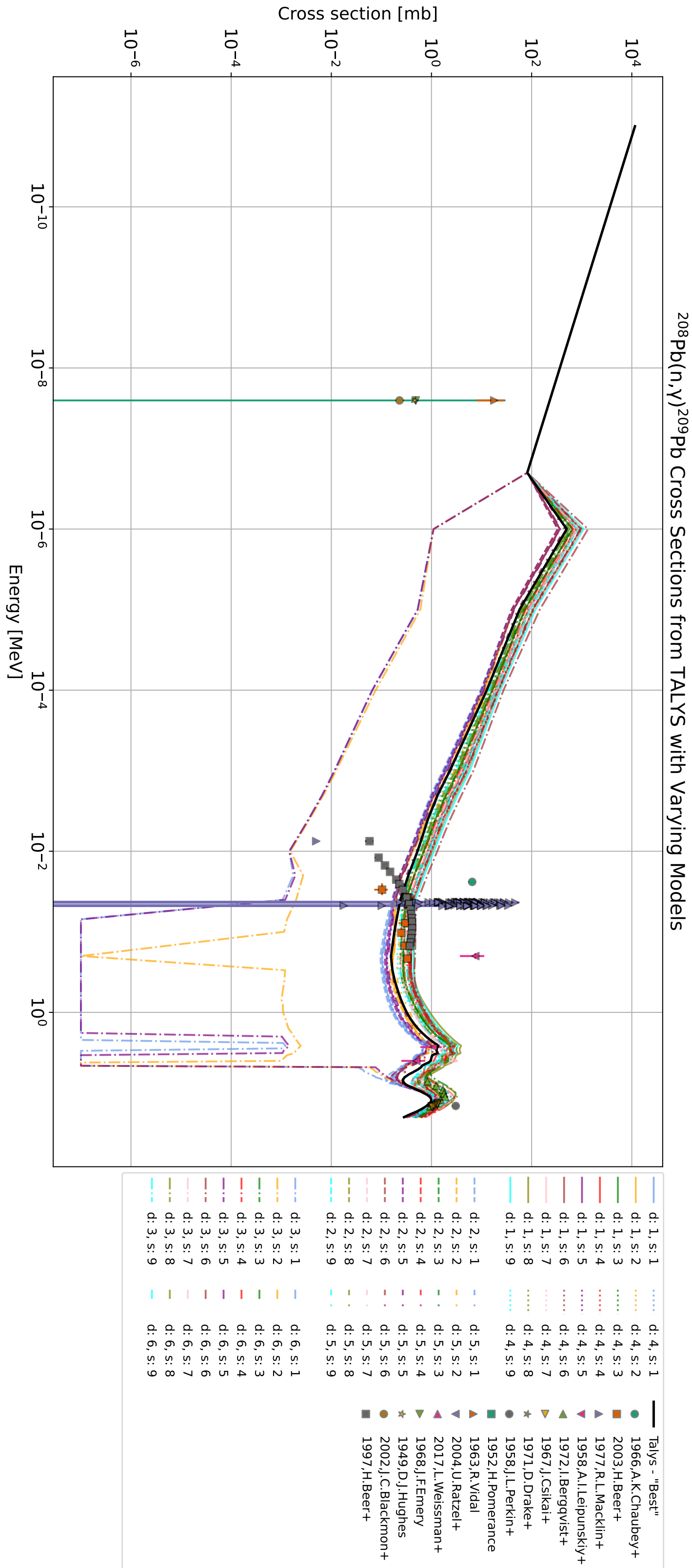

Figure 7: Caption

5.4 Exercise 4

5.4.1 $^{191}\mathrm{Os}(n,\gamma)$ With Experimental γ-Ray Strength Function

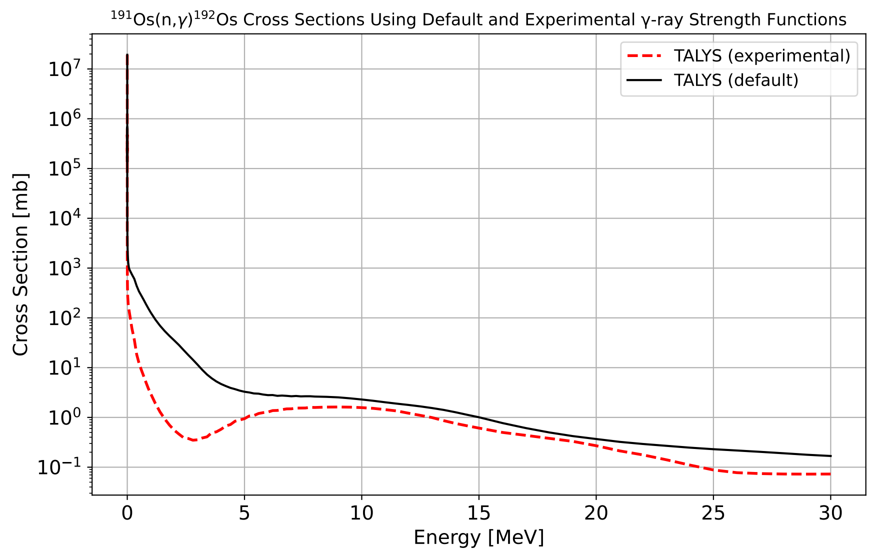

Figure 8: Caption

5.5 Task

5.5.1 $^{120}\mathrm{Sn}(n,\gamma)$

test

6 Conclusion and Outlook

Appendix A: